

1) find the average of numbers from 1 to n.

```
n = int(input("enter the number: "))
```

```
sum = 0
```

```
for i in range(1, n+1, 1):
```

```
    sum += i
```

```
avg = sum//n
```

```
print(avg)
```

n = 5

start

i = 1

stop

i < n+1

condition

~~sum = 0~~
sum += i

step

i = i + 1

i = 1

1 < 6 (True)

sum = 0

sum = sum + i

sum = 1

i = 1 + 1

i = 2

i = 2

2 < 6 (True)

~~sum = 0~~

sum = sum + i

sum = 3

i = 2 + 1

i = 3

i = 3

3 < 6 (True)

~~sum = 0~~

sum = sum + i

sum = 6

i = 3 + 1

i = 4

i = 4

4 < 6 (True)

sum = sum + i

sum = 10

i = 4 + 1

i = 5

i = 5

5 < 6 (True)

sum = sum + i

sum = 15

i = 5 + 1

i = 6

i = 6

6 < 6 (false)

X

X

sum = 15

avg = 15 / 5 = 3

2) Find the sum of squares of numbers from 1 to n

$n = \text{int}(\text{input}(\text{"enter the number: "}))$

sum = 0

for i in range(1, n+1, 1):

sum += ~~i~~ i**2

print

n=5

start

i=i

i=1

i=2

i=3

i=4

i=5

i=6

stop

i < 6

1 < 6 (True)

2 < 6 (True)

3 < 6 (True)

4 < 6 (True)

5 < 6 (True)

6 < 6 (False)

logic condition

sum = i**2

sum = 1**1 = 1

sum += 2**2 = 1+4=5

sum += 3**3 = 5+9=14

sum += 4**4 = 16+14=30

sum = 5**5 = 25+30=55

Step

i=i+1

i=1+1

i=2

i=3

i=3+1

=4

i=4+1

=5

i=5+1

=6

3) Find the sum of cubes of numbers from 1 to n

$n = \text{int}(\text{input}(\text{"enter the number: "}))$

sum = 0

for i in range(1, n+1, 1)

sum += i**3

print(sum)



4) calculate the power of a number without using the $**$ operator

1) $power = \text{int}(\text{input}(\text{"enter the number: "}))$
 $base = \text{int}(\text{input}(\text{"enter the number base: "}))$

$sum = 1$
 $\text{for } i \text{ in range}(1, power + 1, 1):$

$sum *= base$

$\text{print}(sum)$

$sum = 1$
 $base = 2$ $power = 5$

Start

$i = 1$

stop
 $i \neq 6$

$i = 1$

$1 \neq 6$ (True)

$i = 2$

$2 \neq 6$ (True)

$i = 3$

$3 \neq 6$ (True)

$i = 4$

$4 \neq 6$ (True)

$i = 5$

$5 \neq 6$ (True)

$i = 6$

$6 \neq 6$ (False)

condition/logic.

$sum *= base$

$sum = 1$

$sum *= base = 2 \times 1 = 2$

$sum = 2$

$sum *= base = 2 \times 2 = 4$

$sum = 4$

$sum *= base = 4 \times 2 = 8$

$sum = 8$

$sum *= base = 8 \times 2 = 16$

$sum = 16$

$sum *= base = 16 \times 2 = 32$

Step

$i = i + 1$

$i = 1 + 1 = 2$

i

$i = 2 + 1 = 3$

$i = 3 + 1 = 4$

$i = 4 + 1 = 5$

$i = 5 + 1 = 6$

5) Display the first n number of the fibonacci series

n = int(input("Enter the number:"))

~~for~~

sum = 0

pre = 1

for i in range(1, n+1, 1):

print(sum)

a = sum

sum += pre

pre = sum.

~~print~~ /s

n=5

Start

i=1

i=1

i=2

n=5

stop

i < n+1

1 < 6 (True)

2 < 6 (True)

sum=0

pre=1

condition

sum=0

pre=1

print(sum)

a=0

sum += pre $\Rightarrow 0 + 1 = 1$

pre = a

print(sum)=1

a=1

sum += pre

$= 0 + 1 = 1$

pre = a

step

i = i + 1

i = 1 + 1

i = 2

i = 2 + 1

i = 3

$i = 3$

~~3~~ 6 (True)

print(sum) = 1

$i = i + 1$

$a = 1$

$i = 3 + 1$

$sum + pre = 1 + 1 = 2$

$= 4$

$pre = a$

$i = 4$

~~4~~ 6 (True)

print(sum) = 2

$i = i + 1$

$a = 2$

$i = 4 + 1$

$sum = sum + pre = 2 + 1 = 3$

$i = 5$

$pre = a$

$i = 5$

~~5~~ 6 (True)

print(sum) = 3

$i = i + 1$

$a = 3$

$i = 5 + 1$

$sum = sum + pre = 3 + 2 = 5$

$pre = a$

$i = 6$

~~6~~ 6 (True)

X

X

0, 1, 1, 2, 3

output: 0, 1, 1, 2, 3.

6) Display the first n terms of the series.

ex: $n=4$ output = $1, 1/2, 1/3, 1/4$

```
n = int(input("enter a number: "))
```

```
for i in range(1, n+1, 1):
```

```
    if i == 1:
```

```
        print(i)
```

```
    else:
```

```
        print(1/i)
```

```
        print(f"1/{i}" end=" ")
```

Output = $1, 1/2, 1/3, 1/4$

$n=4$

Start

$i=1$

stop

$i \leq 5$

condit/loop.

if $i=1$ (true)

print(i)

step

$i = i + 1$

$i=2$

$2 \leq 5$ (true)

if $i=1$ (true)

print(f"1/{i}")

$i = 2 + 1$

$i = 3$

$i=3$

$3 \leq 5$ (true)

if $i=1$ (false)

print(f"1/{i}")

$i = 3 + 1$

$i = 4$

$i=4$

$4 \leq 5$ (true)

if $i=1$ (false)

print(f"1/{i}")

$i = 4 + 1$

$i = 5$

$i=5$

$5 \leq 5$ (true)

X

✓

7) Display the first n terms of the series.

1, 11, 111, 1111, . . .

```
n = int(input("enter the no. of series"))
sum = "1"
```

```
for i in range(1, n+1, 1):
```

```
    print(int(sum), end=" ")
    sum = str(sum)
    sum += "1"
```

~~for~~

output = 1 11 111 1111 . . .

n = 5

Start

i = 1

Stop

i < 5

condition/loop

sum += "1"

Step

i = i + 1

i = 1

1 < 5 (True)

~~sum = str(sum)~~

print(int(sum)) = 1

i = 1 + 1

sum = str(sum)

i = 2

sum += "1"

i = 2

2 < 5 (True)

print(int(sum)) = 11

i = 2 + 1

sum = str(sum)

i = 3

sum += "1"

i = 3

3 < 5 (True)

print(int(sum)) = 111

i = 3 + 1

sum = str(sum)

i = 4

sum += "1"

i = 4

4 < 5 (True)

print(int(sum)) = 1111

i = 4 + 1

sum = str(sum)

i = 5

sum += "1"

i = 5

5 < 5 (False)

X

X



2) Display the first n terms of the series

1, 3, 9, 27, 81, - - -

```
n = int(input("Enter the number: "))
```

```
sum = 1
```

```
for i in range(1, n+1, 1)
```

```
    print(sum)
```

```
    sum *= 3
```

sum = 1 + 2
sum